ANALYSIS OF PHYSICAL ACTIONS

Download the physical actions EMG data,
Please read through the readme file
Base paper for the assignment
Presentation on the background to EMG signal classification

Task 1: Read and plot the EMG signals in matlab

1. Locate the EMG data in your computer and display the number of folders in the dataset
    a. Useful functions/commands: dir, disp, sprintf
2. List the text files within each of the sub-directories
    a. For both the Normal and Aggressive actions
    b. files(i).name - to get the file name within files array
3. Read and display the size of each signal array in each file
    a. Functions/commands: size
    b. Display size of any signal array of any subject
    c. display size of all signal arrays of all subjects
4. Read and plot a signal from each of the channels
    a. One each for Normal and Aggressive actions
    b. Useful functions/commands: subplot, plot

Task 2: Feature extraction and visualization

1. For each of the subject, read the EMG data for 8 channels
2. Divide the time series array into individual trials
3. There are 15 trials in each array
4. For each of the trial, divide the signal into N_seg no. of segments
5. For each of these segments, extract mean and variance features
6. Concatenate these features from the N_seg segments
7. Concatenate these feature vectors from 8 channels into a ensemble of 8 x N_seg x 2 features
8. Display the feature vectors of any two classes
9. Save the data into a matlab array

Task 3: Label extraction

1. For each of the subjects, for both Normal and Aggressive activities generate the class labels (1 to 20) using the file count in each sub-directory.
2. Concatenate the target vectors into a single vector from Normal and Aggressive activities

Task 4: Classification of pairs of classes and multiple classes

1. Combine the inputs and outputs into single array with 1200 samples with Nf features
2. Nf = no. of features x no. of segments x no. of channels
3. No. of samples = No. of subjects x no. of classes x no. of trials

Analysis Tasks

1. Sample any two - classes from the 20 classes i.e, 120 x Nf data and report the cross-validation classification performance.
2. Repeat the sampling process for multiple classes (2 < C < 20), determine at how many classes the cross-validation performance starts to degrade rapidly i.e., generate a classification accuracy vs no. of classes plot
   Detailed instructions for solving Q2.
   a. For each of the number of classes M = 2 to 19
      i. Generate the N_c_2 list of combos (nchoosek)
      ii. Pick 10 random numbers between 1 and N_c_2 to select 10 combos of classes to be analyzed (randi)
      iii. For each mc = 1:Nmc % No. of Monte Carlo runs
         1. Extract the selected labels for each of selected combo
         2. Extract the corresponding data for cross validation analysis
         3. Using the inputs and outputs for this selection, apply the multi-class kNN algorithm to to determine the cross-validation performance
         4. Store the accuracy results in an array
   b. Plot the heatmap of the accuracy vs the no. of class and the MC runs
   c. Evaluate the average across the MC runs and plot the mean accuracy vs. no. of classes M

Instructions for the report: (Maximum 10 and minimum 3 pages)

A pdf report with the following contents should be uploaded by each student

1. Motivation for the classification problem
2. Introduction to the problem physical actions classification using machine learning approaches
3. A block diagram of the proposed classification approach
4. Implementation of the method on the physical actions dataset (include a very short description of the dataset)
5. Case studies
   a. Binary classification analysis with plots and description of the performance
   b. Multi-class classification analysis with plots and discussion on the results
6. Conclusion

Notes:

1. Please use an IEEE journal paper template for preparing the report - (it can easily found in the Overleaf libraries). Double column format is recommended. Gallery - Templates, Examples and Articles written in LaTeX - Overleaf, Online LaTeX Editor
2. The code for the assignment should be included as an appendix (or can be properly linked) in the report. (Not considered toward the page count)